TARAS

sense . scale . sustain

Smart Agriculture Analysis & Simulation System

TARAS is an intelligent agriculture platform that integrates IoT-based ssensing, digital-twin visualization, and AI-driven irrigation recommendations. The system helps farmers optimize water usage, reduce input costs, and support sustainable field management.

Energy-efficient LoRa communicaiton

Long-range, low-power LoRa ensures stable and reliable data transmission across the field.

Digital twin representation

Real-time 2D/3D moisture maps for smarter decisions.

Carbon footprint suggestion

Estimates carbon emissions using farmer inputs .

AI-assisted irrigation suggestion

ML predicts optimal watering and sends instant alerts for over- or under-irrigation.

Real-time soil sensing

TARAS monitors soil moisture and temperature, giving zone-based real-time insights

Weather-aware decision support

Irrigation recommendations adapt automatically to rainfall, temperature, and evaporation forecasts.

UN Sustainable Development

2 ZERO HUNGER

Supports sustainable agriculture and food security by improving irrigation efficiency and increasing crop yield through data-driven water management.

9 INDUSTRY, INNOVATION AND INFRASTRUCTURE

Promotes agricultural digitalization and innovation by integrating IoT sensors, machine learning, and digital twin technology into a unified smart farming system.

13 CLIMATE ACTION

Contributes to climate-change mitigation by optimizing water and energy use, preventing over-irrigation and reduce environmental impact.

System Overview

Soil moisture & temperature IoT node

LoRa-based gateway (Raspberry Pi)

Digital twin farmland visualization (3D field)

TARAS aims to transform traditional farming into a data-driven practice by providing continuous field monitoring, irrigation optimization, and sustainability analytics. The system supports farmers in making better decisions with reduced resource consumption.